

AJAY TEWARI

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Summary

Staff software engineer with 19 years of experience designing and operating distributed, cloud-native backend systems and microservices on AWS and GCP. Expertise in Python and Java, streaming architectures (Kafka), NoSQL databases, and containerized environments. Proven track record optimizing caching strategies to reduce cloud costs and driving analytics solutions. Skilled in applying AI to enhance conversational and semantic search pipelines.

Experience

Xero

Apr 2026 - Present

Senior Software Engineer (Technical Lead)

San Mateo, CA

- Designed, built, and launched a new subscription service from 0 to 1 within weeks of joining, scaling it to process 34 million events per day as adoption grew.
- Architected and developed AWP Yoda, a suite of tools for end-to-end testing, usage adoption dashboard, regression testing, A/A testing and A/B testing at scale, from the ground up; adopted by multiple consumer teams, with rollout to additional teams underway.
- Built an end-to-end BDD smoke test suite that gates PRs and deployments by verifying the AWP platform works across every environment, catching issues before customers do.
- Built PR Sentry, a multi-model AI code reviewer deployed as a reusable GitHub Action, adopted by Xero repositories as a change-management control for pull requests.

Roku Inc.

Aug 2021 - Mar 2026

Senior Software Engineer (Technical Lead)

San Jose, CA

- Led a team to design and develop conversational agent components powered by generative and agentic AI, including model evaluation pipelines and semantic caching, utilizing end-to-end regression pipelines to ensure robustness.
- Architected and implemented distributed systems and cloud-native microservices using Docker and Kubernetes on AWS, leveraging Python with strong software design principles.
- Led a team to build a multi-turn dialog manager for media and IoT queries, orchestrating agentic workflows that enhanced user experience.
- Implemented caching strategy using Redis to optimize cloud costs, reducing downstream service traffic by 9%.
- Drove improvements in user engagement for voice music queries, achieving a 7% increase.
- Led cross-functional initiative to boost engagement for sports and live channels.

Optimizely, Inc

Oct 2019 - Aug 2021

Senior Software Engineer (Technical Lead)

San Francisco, CA

- Led a team to deliver scalable backend microservices for billing on AWS using Python and Domain-Driven Design, which reduced billing processing latency and improved system reliability
- Designed and launched an Entitlement Management System to handle hundreds of entitlements across diverse products.
- Optimized batch usage counting system to process billions of events from 1500+ enterprise accounts, streamlining evaluation of Monthly Unique Visitors and Impressions.
- Built an automated GDPR compliance platform to efficiently manage customer Subject Access Requests.
- Developed backend APIs for impressions usage counting and billing visualization, enhancing reporting capabilities.
- Championed tech debt reduction initiatives to improve platform performance, scalability, and adaptability, enabling 2x business growth through agile practices.

Oracle, Corporation

Jan 2017 - Oct 2019

Senior Software Engineer

San Jose, CA

- Optimized cost sheet component performance, reducing summary record update performance (O(n) to O(k)).
- Implemented bulk print functionality, reducing business processes documents print time by 80%.
- Redesigned end-to-end line items consolidation, which resulted in 60% latency improvement.
- Designed, built, and unit-tested backend APIs for web-based cost sheet components with Java and MySQL, delivering more reliable data handling for downstream services
- Designed and developed reusable attachment picker component for file uploads.

Exilant, Technologies / Apple Inc.

Apr 2016 - Dec 2016

Technical Lead

Sunnyvale, CA

- Led a team to implement REST-based microservices APIs for Kafka producer/consumer using Docker containers, enabling faster deployment and more reliable message processing

- Optimized Kafka messages consumption rate by 40%.
- Developed log scanning tool to measure business entity metrics.
- Developed analytics dashboard with tree-based representation for Apple Care traffic analytics.

Accenture LLP / Cisco Systems

Jul 2011 - Apr 2016

Technical Lead

San Jose, CA

- Led a team of 8 engineers using Agile and Jira to deliver the order management product from concept through production, meeting all release milestones
- Designed and implemented REST based backend APIs for order management application.
- Delivered build pipeline for 3 applications and optimized build time by 90%.
- Implemented security infrastructure and tooling to protect an enterprise application against XSS/CSRF attacks using OWASP.

Wipro, Technology

Sep 2010 - Jul 2011

Senior Software Engineer

Bengaluru, KA, India

- Designed and implemented web services for retail services.
- Designed and implemented web-based dashboard for web services testing.

CGI Group

Jun 2007 - Sep 2010

Senior Software Engineer

Bengaluru, KA, India

- Implemented and optimized PL/SQL stored procedures for portfolio performance reports for a investment bank.
- Designed and implemented a dashboard for daily investment bank portfolio performance.
- Designed and implemented a web based dashboard for cron jobs scheduling requests.

Education

KIIT University

Aug 2004 - May 2007

Master, Computer Applications

Bhubaneswar, Odisha, India

- **Coursework:** Data Structures, Machine Learning, Internet Technology, Mathematics, Software Eng., Artificial Intelligence, Systems Programming, Operating Systems, Algorithms Analysis, Database Management, Computer Architecture, Networking

Certifications

- **Certification in Machine Learning and Artificial Intelligence:** UC Berkeley · Jan 2025 – Jul 2025
- **AWS Certified Cloud Practitioner:** AWS · Feb 2025

Technical Skills

- **Languages:** Python, TypeScript/JavaScript, Go, Node.js, SQL, Java
- **Artificial Intelligence:** RAG, Agentic AI, Orchestration, Vector Databases & Semantic Search, MCP, Prompt/Context Engineering, Generative AI, AI/ML Systems
- **Cloud Platforms:** AWS, Google Cloud Platform
- **Data & Streaming:** Airflow, Apache Kafka, Redis, Temporal
- **Infrastructure & DevOps/MLOps:** Docker, Terraform, Kubernetes, MLFow, Linux
- **Monitoring & Observability:** Grafana, Datadog
- **Databases:** DynamoDB, Oracle, MySQL, NoSQL Databases
- **Version Control:** GitHub/Gitlab
- **Software Architecture & Methodologies:** Microservices, Unit Testing Frameworks, Agile Development, Software Design, Software Development, Distributed Systems

Recognition and Awards

- **Exemplary Badge:** Exemplary capstone submission from UC Berkeley ML Certification.
- **ACE Team Award (Q1-FY15):** Best performing team award.
- **Outstanding Performer Award:** Client recognition for designing CI build pipeline.
- **Multiple Stellar Awards:** Key contributor recognition for exemplary contributions (Q1-FY12, Q3-FY12).
- **Monetary Awards:** 10+ points based monetary awards for small but impactful wins.

Leadership / Extracurricular

KIIT University

Aug 2004 - May 2007

Founder Member

- The mission of the club was to help each other to improve on our technical and soft skills by organizing various cultural, sports and academic activities.
- Outcome was 200% increase in on-campus job offers compared to previous batch.